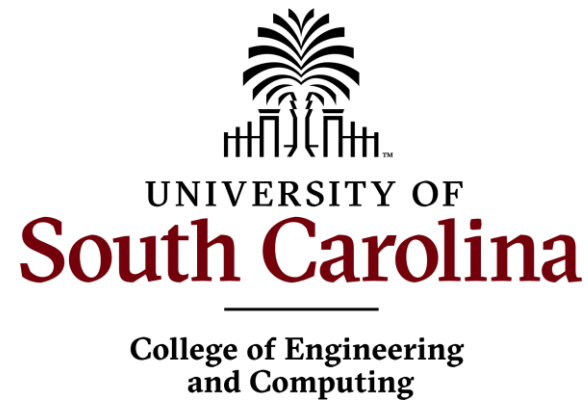


INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: Linear & Quadratic Interpolation

Lang Yuan, Associate Professor

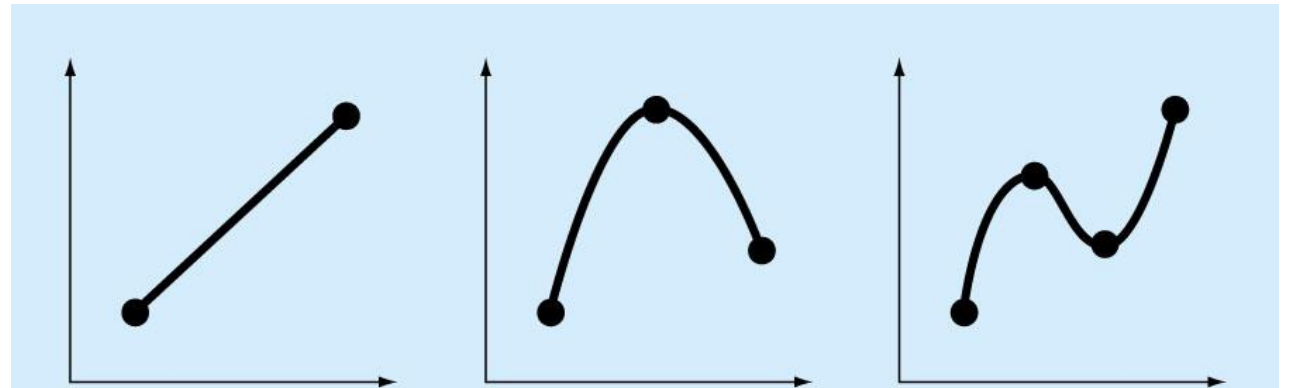


INTRODUCTION

- Estimation of intermediate values between precise data points. The most common method is:

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

- Although there is one and only one n^{th} -order polynomial that fits $n + 1$ points, there are a variety of mathematical formats in which this polynomial can be expressed:
 - The Newton polynomial
 - The Lagrange polynomial



**(a) 1st-order
(linear)
connecting 2 points** **(b) 2nd-order
(quadratic)
connecting 3 points** **(c) 3rd-order
(cubic)
connecting 4 points.**

LINEAR INTERPOLATION

Linear Interpolation

- The interpolated value lies on a straight line connecting two data points $(x_0, f(x_0)), (x_1, f(x_1))$

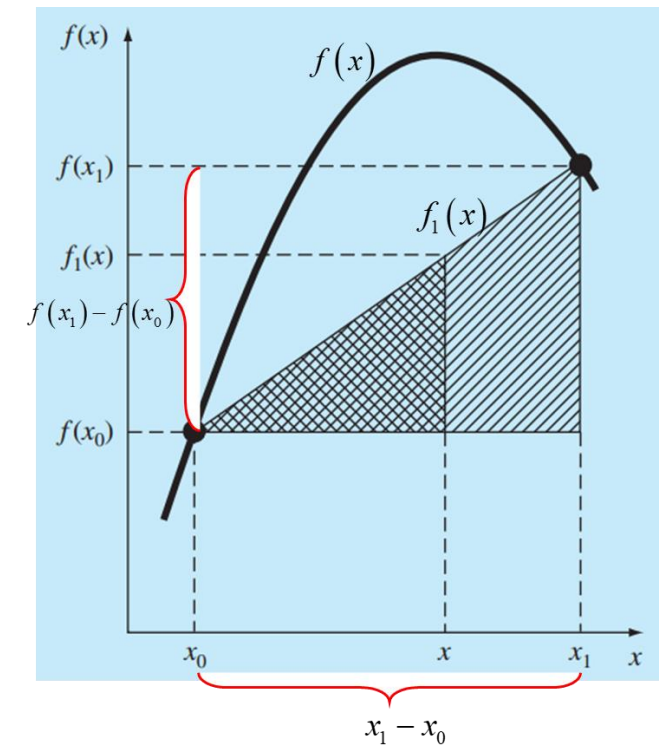
Interpolated value @ x

$$\frac{f_1(x) - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Subscript "1": First-order (Linear) interpolating polynomial for " f "

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$$

Finite-divided-difference approximation
of the first derivative



The smaller the interval between data points, the better the approximation!

AN EXAMPLE

Estimate the natural logarithm of $x = 2$ using linear interpolation

- (a) Interpolating between two data points $x_0 = 1$ and $x_1 = 6$
 - (b) Repeat the interpolation between two data points $x_0 = 1$ and $x_1 = 4$
- (True value of $\ln(2) = 0.6931472$)

Solution:

First obtain $f(x_0) = \ln(1) = 0$ and $f(x_1) = \ln(6) = 1.791759$

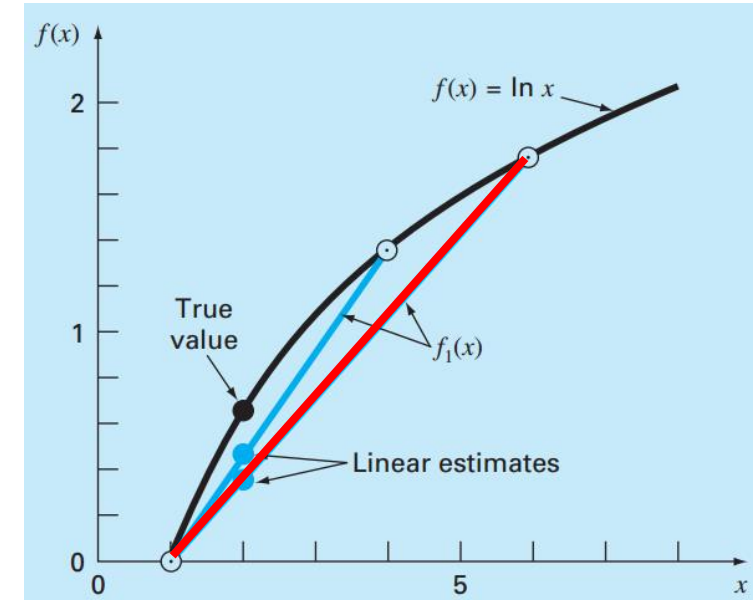
$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$



$$f_1(2) = f(1) + \frac{f(6) - f(1)}{6 - 1}(2 - 1) = 0 + \frac{1.791759 - 0}{6 - 1} = 0.3583519$$



$$\varepsilon_t = \left| \frac{f_t - f_1}{f_t} \right| \times 100\% = \left| \frac{0.6931472 - 0.3583519}{0.6931472} \right| \times 100\% = 48.3\%$$



AN EXAMPLE

Estimate the natural logarithm of $x = 2$ using linear interpolation

- (a) Interpolating between two data points $x_0 = 1$ and $x_1 = 6$
 - (b) Repeat the interpolation between two data points $x_0 = 1$ and $x_1 = 4$
- (True value of $\ln(2) = 0.6931472$)

Solution:

First obtain $f(x_0) = \ln(1) = 0$ and $f(x_1) = \ln(4) = 1.386294$

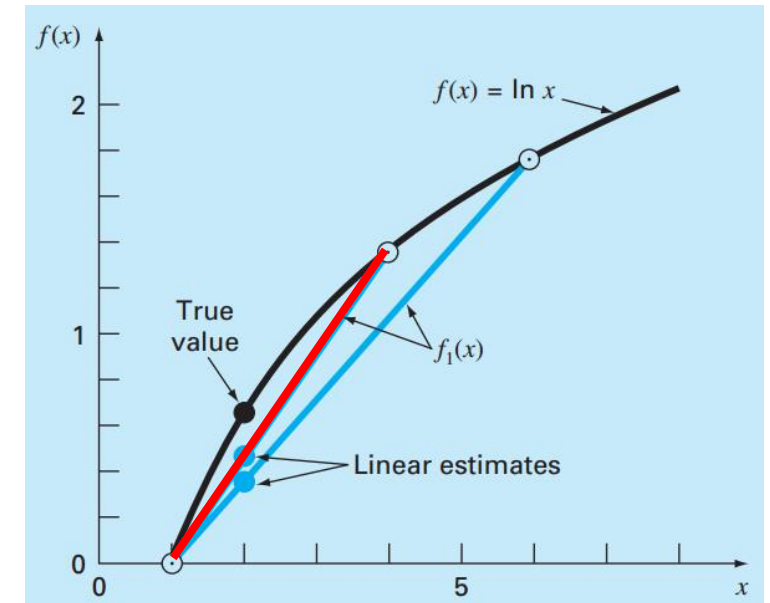
$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$



$$f_1(2) = f(1) + \frac{f(4) - f(1)}{4 - 1}(2 - 1) = 0 + \frac{1.386294 - 0}{4 - 1} = 0.4620981$$



$$\varepsilon_t = \left| \frac{f_t - f_1}{f_t} \right| \times 100\% = \left| \frac{0.6931472 - 0.4620981}{0.6931472} \right| \times 100\% = 33.3\%$$



QUADRATIC INTERPOLATION

Quadratic Interpolation

- Introduce some curvature into the line connecting the points
- A 2nd-order polynomial (quadratic polynomial): three data points are available

Subscript "2": 2nd-order (quadratic)
interpolating polynomial for "f"

2nd order curvature

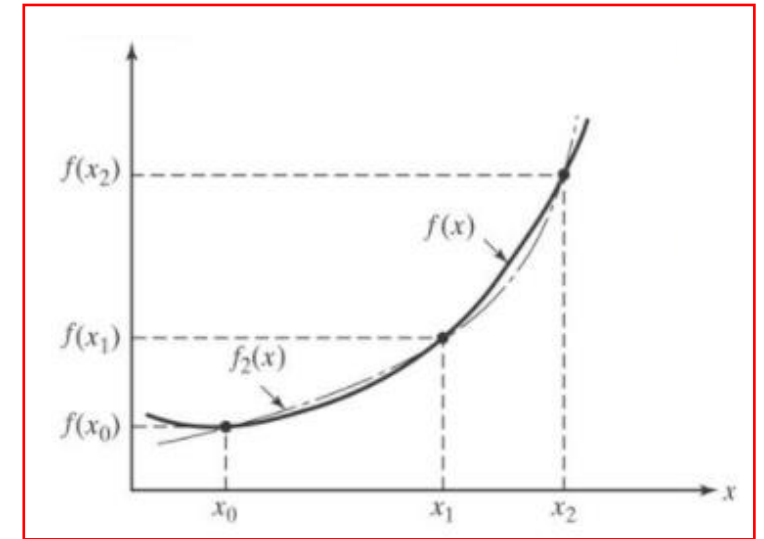
$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$@ x = x_0, b_0 = f(x_0)$$

$$@ x = x_1, b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Approximated first derivative/Slope

$$@ x = x_2, b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$



AN EXAMPLE

Estimate the natural logarithm of $x = 2$ using quadratic interpolation and three data points

(True value of $\ln(2) = 0.6931472$)

Solution:

$$\begin{array}{ll} x_0 = 1 & f(x_0) = 0 \\ x_1 = 4 & f(x_1) = 1.386294 \\ x_2 = 6 & f(x_2) = 1.791759 \end{array}$$

$$b_0 = f(x_0) = 0$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1.386294 - 0}{4 - 1} = 0.4620981$$

$$b_2 = \frac{\frac{1.791759 - 1.386294}{6 - 4} - \frac{1.386294 - 0}{4 - 1}}{6 - 1} = -0.0518731$$

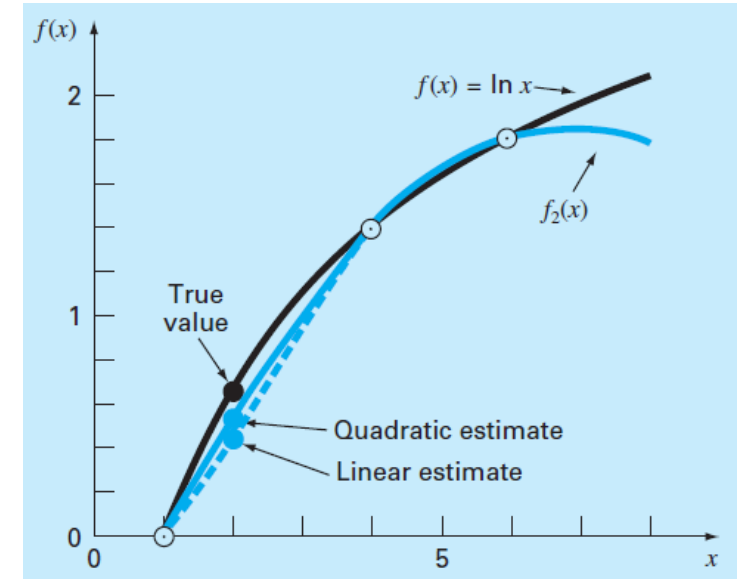
$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$f_2(x) = 0 + 0.4620981(x - 1) - 0.0518731(x - 1)(x - 4)$$

$$f_2(x = 2)$$

$$f_2(2) = 0.5658444$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$



$$\varepsilon_t = \left| \frac{f_t - f_l}{f_t} \right| \times 100\% = \left| \frac{0.6931472 - 0.5658444}{0.6931472} \right| \times 100\% = 18.4\%$$